```python
import numpy as np
import matplotlib.pyplot as plt

from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

# XOR dataset
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

y = np.array([0, 1, 1, 0])

# Neural Network
model = MLPClassifier(
    hidden_layer_sizes=(4,),
    activation='relu',
    solver='adam',
    learning_rate_init=0.1,
    max_iter=2000,
    random_state=42
)

# Train
model.fit(X, y)

# Prediction
prediction = model.predict(X)

print("Actual Output:")
print(y)

print("\nPredicted Output:")
print(prediction)

print("\nAccuracy:")
print(accuracy_score(y, prediction))
```

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    random_state=42
)

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print(prediction)

print("\nAccuracy:")
print(accuracy_score(y, prediction))

# Loss graph
plt.figure(figsize=(8, 5))

plt.plot(
    model.loss_curve_,
    marker='o'
)

plt.xlabel("Iterations")
plt.ylabel("Loss")
plt.title("Backpropagation Training Loss")
plt.grid(True)

plt.show()
```

```
Actual Output:
[0 1 1 0]

Predicted Output:
[1 1 1 1]

Accuracy:
0.5
```

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Predicted Output:
[1 1 1 1]

Accuracy:
0.5
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